

Rational-point undecidability for smooth projective varieties over the rational numbers

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Abstract

Combining OpenAI’s negative solution of rational Hilbert’s tenth problem with Poonen’s established effective transfer, we obtain undecidability of rational-point existence for arbitrary smooth projective geometrically integral varieties over $\mathbb{Q}$. We specify a homogeneous-equation input promise and reconstruct the terminating contradiction algorithm. There is also no computable uniform bound on the height of one rational point in terms of the size of the presented equations. The arithmetic input is used with two explicit corrections documented here and in the accompanying verification supplement. This is a geometric consequence of that input; neither the geometric reduction nor the base undecidability theorem is claimed as a new independent solution.

1 Inputs and precise scope

Let $\text{H10}(\mathbb{Q})$ denote the decision problem with input an integer $n \geq 0$ and a polynomial $f \in \mathbb{Z}[x_1, \dots, x_n]$, asking whether $f(\mathbf{a}) = 0$ for some $\mathbf{a} \in \mathbb{Q}^n$. The number of variables is part of the input. Constants, including $f = 0$, are allowed.

A projective presentation e consists of an integer $N \geq 0$ and a finite list of homogeneous integer polynomials $F_1, \dots, F_r$ in $x_0, \dots, x_N$. It specifies

$$X_e = \text{Proj}(\mathbb{Q}[x_0, \dots, x_N]/(F_1, \dots, F_r)).$$

Define $\text{SPGI}(\mathbb{Q})$ to ask whether $X_e(\mathbb{Q}) \neq \emptyset$, under the promise that X_e is smooth and geometrically integral. Projectivity is supplied by the presentation. Geometric integrality includes nonemptiness of the geometric scheme. There is no bound on N , dimension, degrees, the number of equations, or coefficient heights. The oracle need not terminate on a presentation violating its promise.

The geometric construction and the decision transfer below are due to Poonen [1]. His Remark 1.2(c) credits a prior $\mathbb{Q}$ -case route to R. Robinson, through Smoryński, *Logical Number Theory I*, §II.7. We do not claim a new reduction. Family 004 [2] asserts the arithmetic undecidability input that removes the hypothesis from this established implication. Section 4 specifies the exact input used, its two proof corrections, and the scope of its source audit. The main contribution of this note is an explicit consequence with valid oracle queries and presentation conventions, together with the documented corrections. We make no assertion of first priority for the consequence.

2 Poonen’s effective transfer and the promise

We use the following established input from Poonen’s Theorem 1.3, specialized to $\mathbb{Q}$: given a projective $\mathbb{Q}$ -variety X and an open $U \subseteq X$, one can algorithmically construct a regular

projective variety Y with a morphism $\pi : Y \rightarrow X$ whose image on rational points is exactly $U(\mathbb{Q})$. The output need not be connected or geometrically integral.

The effectivity clause matters. In Poonen's §9, each permissible parameter c gives an output with $\pi_c(Y_c(\mathbb{Q})) \subseteq U(\mathbb{Q})$. An effective cardinality bound on the exceptional parameters allows a finite list containing a parameter for which equality holds. Taking the disjoint union of the outputs gives equality without identifying that good parameter. The boundary exclusion comes from his Châtelet-surface construction. Resolution preserves the lifts on the smooth fibers used in the argument. These are parts of the cited effective theorem, not a claim that naive projective closure preserves rational points.

Lemma 1. *If a connected smooth projective $\mathbb{Q}$ -scheme C has a rational point, then it is geometrically integral.*

Proof. A regular Noetherian scheme has disjoint irreducible components: at an intersection its local ring would have more than one minimal prime, whereas a regular local ring is a domain. Its finitely many components are therefore open and closed. Connectedness implies that C is integral. The finite algebra $L = H^0(C, \mathcal{O}_C)$ is consequently a field over $\mathbb{Q}$. A rational point induces a $\mathbb{Q}$ -algebra map $L \rightarrow \mathbb{Q}$, forcing $L = \mathbb{Q}$. Global functions commute with extension of the base field here. Over $\overline{\mathbb{Q}}$, each smooth connected projective component has only constant functions and is integral. Thus the number of these components is $\dim_{\mathbb{Q}} L = 1$. Smoothness also gives geometric reducedness, proving geometric integrality. This is the $\mathbb{Q}$ -projective instance of Poonen's Lemma 10.1. $\square$

The component decomposition and geometric-integrality tests used below are effective algebraic operations, as in Poonen's proof of Theorem 1.1(i). For clarity, for a smooth projective Y , effective coherent cohomology computes the finite *étale* algebra $H^0(Y, \mathcal{O}_Y)$. Its field factors give the connected components, and a connected component is geometrically integral precisely when its global-functions field has $\mathbb{Q}$ -dimension one. Projective constructions provide embeddings, and elimination gives homogeneous-equation presentations. Regularity and smoothness agree for finite-type schemes over the perfect field $\mathbb{Q}$. These conventions ensure that no invalid input is passed to the oracle.

Theorem 2. *An algorithm deciding $\text{SPGI}(\mathbb{Q})$ on every valid input would decide $\text{H10}(\mathbb{Q})$. Conversely, decidability of $\text{H10}(\mathbb{Q})$ would decide $\text{SPGI}(\mathbb{Q})$.*

Proof. For the forward implication let D be the promised decision procedure. On input f , handle constant polynomials directly. Otherwise form the reduced affine zero scheme U of f and its projective closure X , recording U as an open in X . If U is empty, return NO. Apply Poonen's effective theorem to obtain (Y, π) . Compute the connected components $C_1, \dots, C_s$ of Y and their projective presentations. Test geometric integrality and discard each component that fails. By Lemma 1, discarded components have no rational points. Every retained component is smooth, projective and geometrically integral. Query D on each retained presentation, and return YES if and only if some query answers YES.

Every preprocessing step terminates. The list of valid queries is finite, so D 's promise ensures termination of the whole algorithm. Correctness follows from

$$f(\mathbb{Q}^n) \ni 0 \iff U(\mathbb{Q}) \neq \emptyset \iff Y(\mathbb{Q}) \neq \emptyset \iff \exists i : C_i(\mathbb{Q}) \neq \emptyset.$$

The last condition is exactly the disjunction of the retained oracle answers. If there are no retained components it is false.

Conversely, on a projective presentation take the $N + 1$ standard affine charts, setting $x_j = 1$ on chart j . Their equations have a common rational zero precisely when the sum of their squares does, since $\mathbb{Q}$ is ordered. Call an $\text{H10}(\mathbb{Q})$ decision procedure on these finitely

many polynomials and take the disjunction. Every rational projective point lies in one of these charts, proving the reverse implication. $\square$

The forward construction yields a finite nonadaptive disjunctive oracle reduction, and in particular a Turing reduction. It does not establish a single-output many-one reduction into the promised geometrically integral class. No quantitative or dimension-preservation assertion is used.

Corollary 3. *No algorithm decides rational-point existence for every valid presentation in the class $\text{SPGI}(\mathbb{Q})$.*

Proof. OpenAI's Theorem 1.1 [2], with the proof corrections recorded in Section 4, gives undecidability of $\text{H10}(\mathbb{Q})$ with arbitrary input number of variables. Theorem 2 would contradict it if the stated algorithm existed. $\square$

3 An explicit height consequence

Specify a size parameter as follows. For $a \geq 0$, let $b(a)$ be its binary representation, with $b(0) = 0$, and let $\ell(a)$ be its length. Use the self-delimiting integer code $c(a) = 1^{\ell(a)}0b(a)$. Encode N , r , then the number of monomials in each polynomial. For every monomial encode a sign bit, c of the nonzero coefficient magnitude, and c of each of its $N + 1$ exponents. Require increasing lexicographic exponent vectors and a common total degree within each polynomial. The zero polynomial has zero monomials. These conventions specify a uniquely parsable finite string e ; let $s(e)$ be its bit length. The size refers to this presentation and embedding, not to the isomorphism class of the abstract variety. The general connection between bounding the least height of a rational point and rational Hilbert's tenth problem is prior work; see Skorobogatov–Sofos [5], Section 1.2.

For $P \in \mathbf{P}^N(\mathbb{Q})$ choose its primitive integer coordinates $(z_0, \dots, z_N)$ with the first nonzero coordinate positive, and set $H(P) = \max_j |z_j|$.

Corollary 4. *There is no total computable function $B : \mathbb{N} \rightarrow \mathbb{N}$ such that for every valid e with $X_e(\mathbb{Q}) \neq \emptyset$ there exists $P \in X_e(\mathbb{Q})$ satisfying $H(P) \leq B(s(e))$.*

Proof. Compute $M = B(s(e))$. Enumerate all nonzero primitive integer vectors in $[-M, M]^{N+1}$ with first nonzero coordinate positive, and evaluate the F_i exactly. Return YES if any vector satisfies every equation, and NO otherwise. The enumeration is finite. The hypothesized bound puts a point in the enumeration whenever one exists. This would decide $\text{SPGI}(\mathbb{Q})$, contradicting Corollary 3. For $M = 0$ the enumeration is empty, consistently with $H(P) \geq 1$. $\square$

The proof also rules out a presentation-dependent computable bound that terminates on every valid input. It concerns the height of *one* point when points exist. It makes no assertion about a fixed-dimensional family, finite rational-point sets, or all points on one variety.

4 Arithmetic input and proof corrections

The external arithmetic theorem used is precisely Theorem 1.1 of [2]: no algorithm decides the existence of a rational zero of an arbitrary integer polynomial, with the number of variables part of the input. Its proof is not a geometric argument. A hypothetical rational solvability algorithm decides each finite existential polynomial test in an effectively enumerated ring theory. Disequalities are replaced by inverse variables, and finite systems by sums

of squares. If all tests succeed, compactness gives a term-generated ring in a nonstandard rational field, together with an injective additive elliptic index taking 1 to 1. Pole parity and a five-point height estimate bound the distinguished indices by ordinary integers, so the ring's distinguished roots are ordinary integers. An elementary diagram used to justify compactness is not queried by the algorithm. Dovetailing an integer-zero search with the finite-test search would decide integral Hilbert's tenth problem, yielding the contradiction. The source-body reconstruction and exact dependency checks are included in the verification supplement; finite computation alone is not this proof.

The pole-parity application uses only the non-CM, full rational two-torsion branch of the new pointwise converse [3]. For the curves

$$E_l : y^2 = x(x-l)(x+3l),$$

with l restricted to the parameters selected by family 004's descent, that construction gives $\dim_{\mathbf{F}_2} \text{Sel}_2(E_l/\mathbb{Q}) = 3$ and full rational two-torsion. Thus, with $r = \text{rank } E_l(\mathbb{Q})$,

$$r + \dim_{\mathbf{F}_2} \text{Sha}(E_l/\mathbb{Q})[2] = 1.$$

The extra Selmer class is realized by a rational point using the converse's finiteness conclusion for $\text{Sha}(E_l/\mathbb{Q})[2^\infty]$: the alternating perfect Cassels–Tate pairing forces even dimension of $\text{Sha}[2]$, yielding $r = 1$ and $\text{Sha}[2] = 0$. Low Selmer corank alone does not supply that finiteness. Indeed, the full 2^∞ -Selmer corank is at most one, since the divisible corank of $\text{Sha}[2^\infty]$ is at most $\dim_{\mathbf{F}_2} \text{Sha}[2]$. This is the exact hypothesis of the converse. Each E_l is fixed before its internal auxiliary twists are chosen; constants may depend on E_l . The internal odd twist discriminant must not be confused with $l \equiv 3 \pmod{4}$, whose quadratic field has even discriminant. No CM branch or arbitrary rational-two-torsion case is needed.

Two technical substitutions are made in the proofs we use:

1. For the non-CM five-point height argument of [2], use coefficient primes (5, 2). Published Pan modularity at 5 [4] applies without a residual-image restriction; irreducibility, oddness, potential semistability and distinct Hodge–Tate weights are verified from the surface's Tate module. Prime-independent traces identify the same weight-two form at 2, which supplies its conductor at 5. The modular factor and isogeny-height deductions then give the required power bound. The standalone height supplement details every step and explicitly states its retained geometric inputs. Its five branch values come from the fixed discriminant-210 quotient, not arbitrary five points.
2. In the non-CM cyclotomic determinant of [3], reconcile inverse Frobenius with the direct Euler convention by the universal group-ring correction below. It is integral before character evaluation, has central value one, and introduces no factor of two per auxiliary prime. The real-place Tate error, smoothing denominators and fixed-base clearing are still bounded independently of the number of auxiliary primes.

The supplied component reconstructions and integration audit check their joint use: actual Selmer-cone maps and homotopies, outer specializations, weighted theta traces, the nonvanishing estimates, and bounded central clearing all refer to the same finite arithmetic constructions. Their precise scopes and primary-source hypotheses are recorded in the supplement. No applicable complete Lean formalization of this deduction is claimed.

Universal Euler correction. Write $\Lambda = \mathbb{Z}_2[[t]]$, let O be the completed localization at (2), and let G be the finite elementary two-group in that construction. For an auxiliary odd prime q , let $g_q \in G$ mark its active character and let $Z_q = \theta_q \gamma_q$, where $\theta_q^2 = 1$ and

γ_q is the cyclotomic group-like unit. For the local inertia-cohomology module $V_h(-1)$, the determinant is $P_q(Z) = 1 - (a_q/q)Z + (1/q)Z^2$. Put

$$r_q = \frac{P_q(Z_q)}{Z_q^2 P_q(Z_q^{-1})}, \quad U_q = 1 + \frac{r_q - 1}{2}(1 + g_q).$$

The Euler denominators are $O[G]$ -units in the construction. The identity $P_q(Z) - Z^2 P_q(Z^{-1}) = ((q-1)/q)(1 - Z^2)$ gives $r_q - 1 \in 2O[G]$. At an unused character ($g_q = 1$), $U_q = r_q$; at an active character ($g_q = -1$), $U_q = 1$. Multiplying the inverse-convention universal class by $\prod_q \gamma_q^2 \prod_q U_q$ therefore gives the required direct Euler factor at every unused prime and a unit γ_q^2 at each active one. These latter units specialize to one at $t = 0$. For a smoothing integer c , set $A_c^\pm = c^2 - c\gamma_c^{\pm 1}$ and $R_c = A_c^+ / (\gamma_c A_c^-)$. Since $A_c^+ - \gamma_c A_c^- = c(c+1)(1 - \gamma_c)$, R_c is an integral unit congruent to one modulo two, with $R_c(0) = 1$. Starting with the inverse-convention *smoothed* class z_{raw}^- , multiply by $\gamma_c \gamma_d R_c R_d \prod_q \gamma_q^2 U_q$ before dividing by $A_c^+ A_d^+$. This equals $(\prod_q \gamma_q^2 U_q) z_{\text{raw}}^- / (A_c^- A_d^-)$. It imposes the direct smoothing convention without changing central valuations. These are operations on the universal class, not divisions by character idempotents. The full Kato normalization and determinant calculation are in the supplement.

5 Boundaries, attribution and verification

Naive compactification cannot replace the effective transfer: $\text{Spec } \mathbb{Q}[x, y]/(x^2 + 1)$ has no rational point, whereas its closure $V(X^2 + Z^2) \subseteq \mathbf{P}^2$ contains $[0 : 1 : 0]$. Resolution alone need not preserve point existence in the other direction: $V(x^2 + y^2)$ has a rational singular point but its normalization is $\mathbf{A}_{\mathbb{Q}(i)}^1$, which has no rational point. A morphism from a resolution cannot create a rational point when its target has none. The issue is losing singular points or retaining boundary points introduced by compactification.

No conclusion for curves, surfaces, fixed dimension, Fano varieties, general type, or a finite-point promise follows automatically. This note asserts neither a Turing degree nor quantitative bounds beyond the stated oracle implication. The height corollary is an elementary computability consequence and adds no independent arithmetic input.

The geometric implication was already established by Poonen, with a prior rational-field route credited in his Remark 1.2(c). The arithmetic breakthrough is attributed to OpenAI's manuscript-specific citation. The consequence and its elementary height corollary are not advertised as independent solutions of rational Hilbert's tenth problem or as a new geometric reduction. The accompanying priority audit records public source dates separately from manuscript dates and search limitations.

Disclosure. AI tools were used extensively in research, drafting and adversarial verification. Those automated reviews are not conventional human peer review. The preprint has not undergone conventional human peer review or refereeing at publication. The verification package supplies source identities, checkable derivations, independent automated review reports and exact finite checks; it does not claim formal verification of the full theorem.

References

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